Cooling time through a flat plate

Which method predicts the hottest-point threshold accurately at the lowest measured cost?

Synthetic input	Baseline	Units
Full thickness L	0.010	m
Thermal diffusivity	1.0×10^{-5}	m^2/s
Initial / face / target	100 / 20 / 40	°C

Outputs

Temperature profiles
Hottest-point cooling curve
First threshold-crossing time

Ideal controlled face temperatures and constant properties. Preliminary numerical study.

Explicit and implicit finite differences

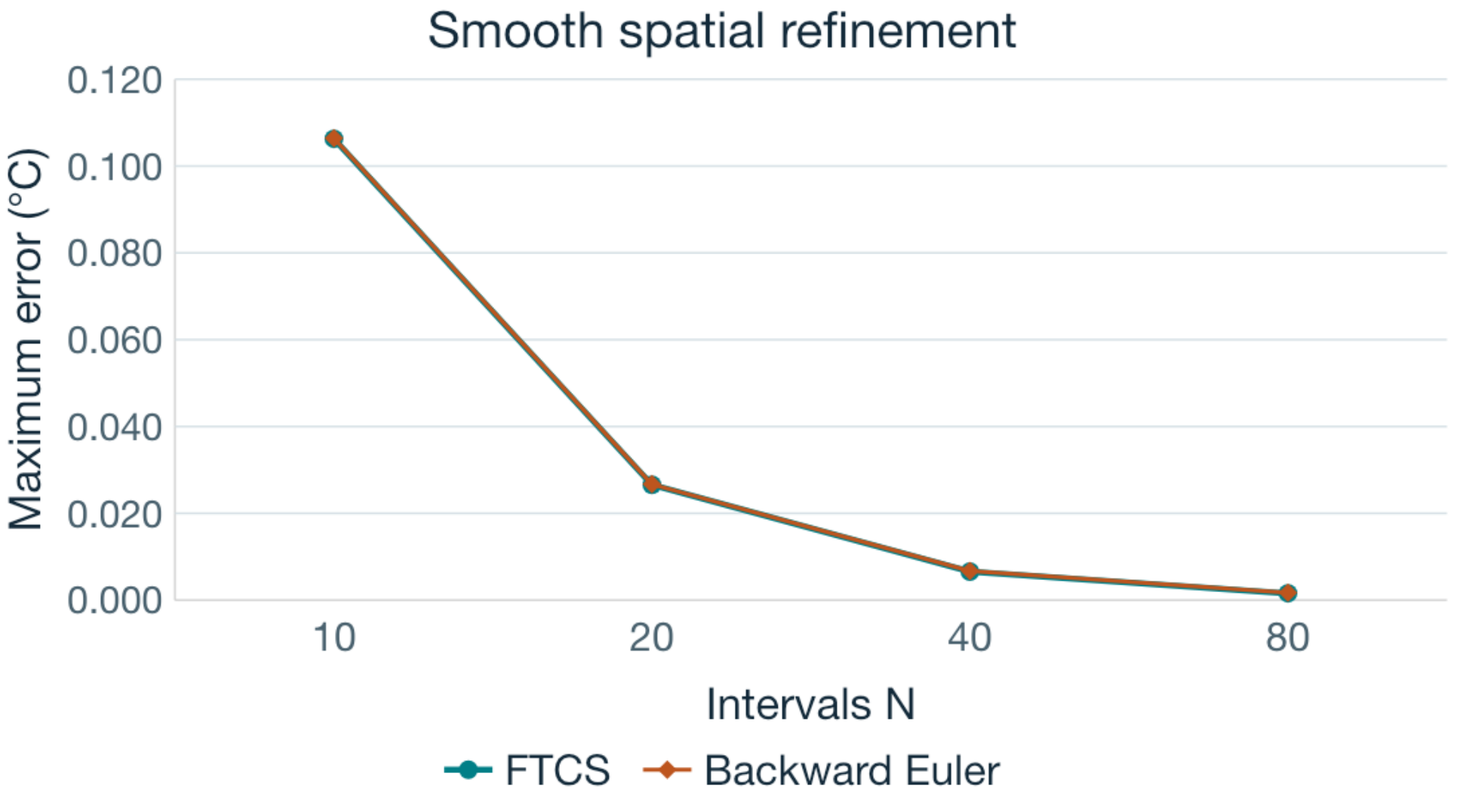
Heat equation: $T_t = \alpha T_{xx}$

FTCS	Backward Euler
Update from the old field	Solve a tridiagonal system
$r = \alpha \Delta t / \Delta x^2 \leq 0.5$	Stable for any positive step
First order in time	First order in time
Second order in space	Second order in space

Accuracy still limits the implicit time step

One-dimensional conduction. Equal fixed face temperatures. No heat source, phase change or convection model.

Verification against exact solutions



Two analytical checks

Smooth sine solution
Odd-sine cooling series

Reference crossing
1.649365 s

Spatial order ≈ 2
Time order ≈ 1

Smooth sine at t = 0.2 s. Fixed dt = 0.00001 s. Separate temporal check uses the exact semidiscrete mode.

Method selection from measured evidence

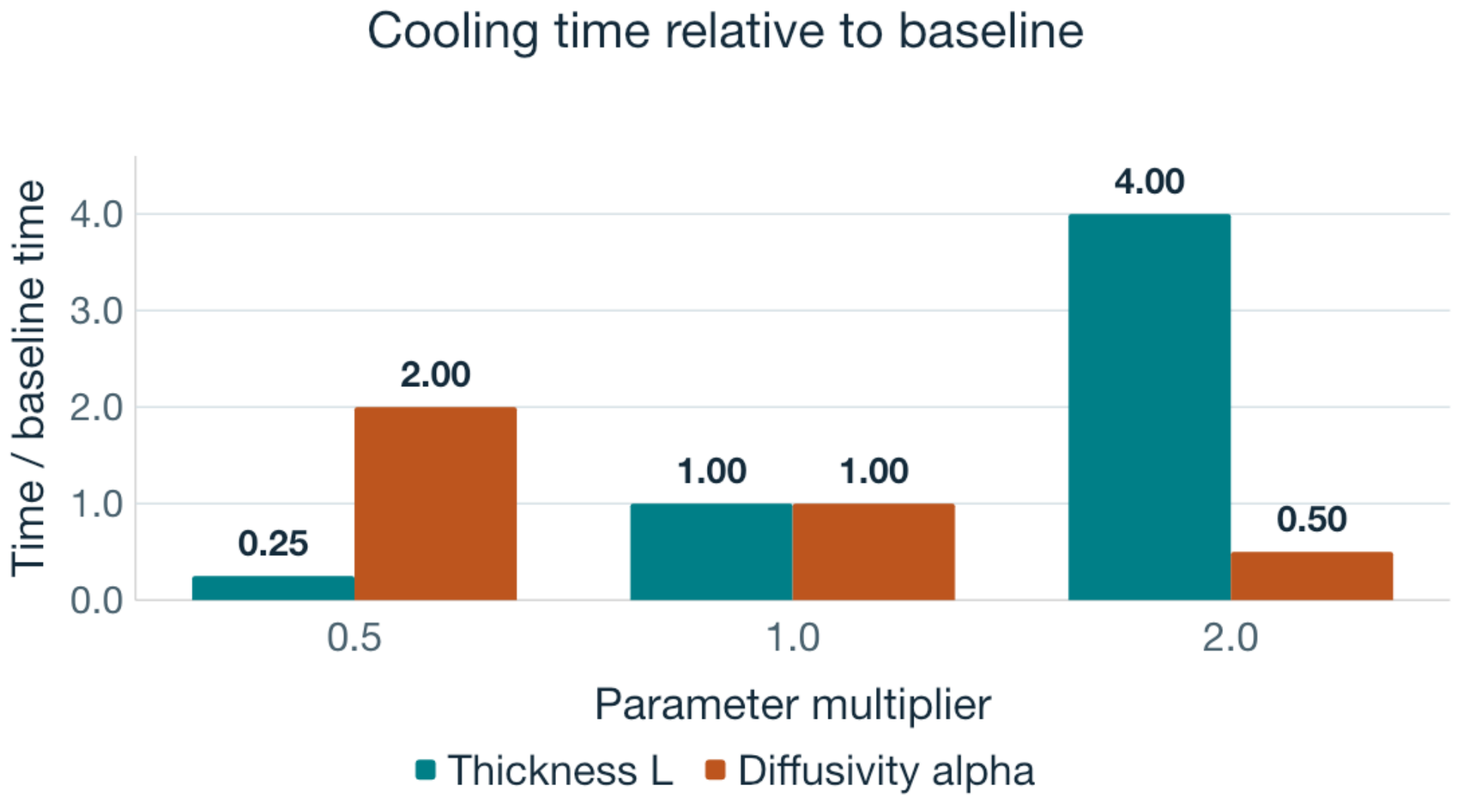
Method	N	dt (s)	Max error (°C)	Crossing error (%)	Median (ms)
FTCS	20	0.00625	0.589	0.229	2.873
Backward Euler	40	0.00625	0.709	0.328	6.494

Selected: FTCS, N = 20, dt = 0.00625 s

Initial transfer case: 1.180 °C failure. Revised setting: fresh confirmation 0.552 °C, 0.214% crossing error.

Selected solve 11.65× faster than the fine baseline. Separate vectorisation check: 8.05×, identical output.

Thickness and diffusivity sensitivity



t scales with L^2 / alpha

Double thickness
4 × cooling time

Double diffusivity
0.5 × cooling time

Analytical verification supports the model calculation. Real-process validation needs measured properties and boundary data.

Repository demonstration and AI disclosure

Live walkthrough

README and reproducible commands
Source and generated API documentation
Tests, report and saved results
Genuine development history

Substantial AI contribution

Codex authored substantial code, tests, analysis and report drafts.

Student review and independent understanding remain essential.

The final setting meets the checked numerical targets. Real plate predictions still need experimental evidence.